

The demo has four preset buttons at the bottom-left. The script walks them in a specific order and expects specific things on screen. This is the mapping — one page per preset, screenshot on the left, exactly what to say on the right, and the number to point at.

ORDER	SCRIPT §	TIME	CLICK	WHAT LANDS	THE NUMBER
1	§3.1 – 3.2	45 s + 45 s	HEALTHY	Baseline. Envelope spectrum is grass. Nothing at 107 Hz. Normal lamp.	≈ 0.3
2	§3.3	2.5 min ← key moment	EARLY DEFECT	22% spall. Fault invisible in raw spectrum, obvious in envelope at 107.4 Hz. Alert lamp.	≈ 3
3	§3.5 first half	30 s	ADVANCED	78% severity. Visible everywhere including raw waveform. Fault lamp.	≈ 7–8
4	§3.5 second half	2 min	NOISY PLANT	Same 22% defect. Noise raised. Index falls, annunciator returns to NORMAL. False negative.	≈ 1.6

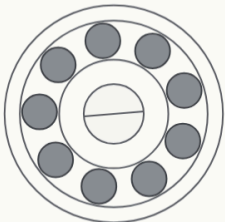
§3.3 — Early defect — is the pedagogical moment of the entire lecture. Click Early defect, point at the raw spectrum, ask the room "tell me where the fault is." Wait. Let them fail to find it. Then show the envelope spectrum where the peak at 107.4 Hz is unmistakable. Then the number: index 0.3 → 3, annunciator Normal → Alert. Same data. Same measurement. The only difference is envelope analysis.

Bearing Fault Lab

SKF 6205 deep-groove ball bearing, 9 rolling elements. Raise the outer-race defect and watch where it shows up: barely in the raw spectrum, unmistakably in the envelope spectrum.

Normal Alert Fault

BEARING - OUTER RACE DEFECT AT 12 O'CLOCK



DEFECT PASS RATE = BPFO = 107.4 Hz ORBIT SLOWED 17x

ROLLING ELEMENT DEFECT STRIKE

OPERATING CONDITIONS

SHAFT SPEED 1797 rpm

DEFECT SEVERITY 0 %

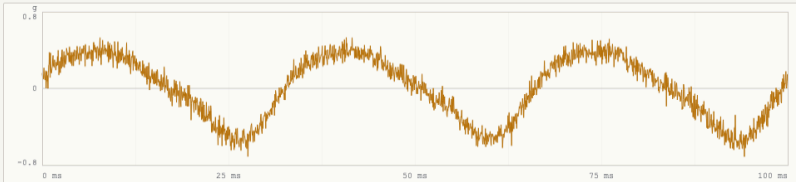
BACKGROUND NOISE 18 %

SHAFT IMBALANCE 35 %

HEALTHY EARLY DEFECT ADVANCED

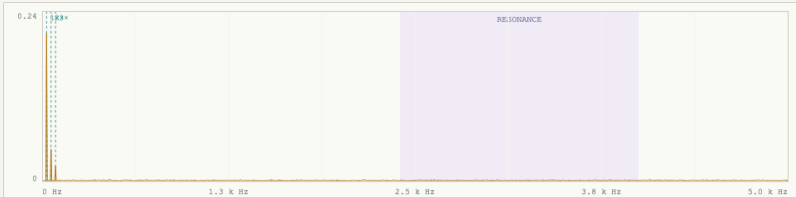
NOISY PLANT

1 - TIME WAVEFORM, RAW ACCELEROMETER ($F_0 = 20\,460\text{ Hz}$)



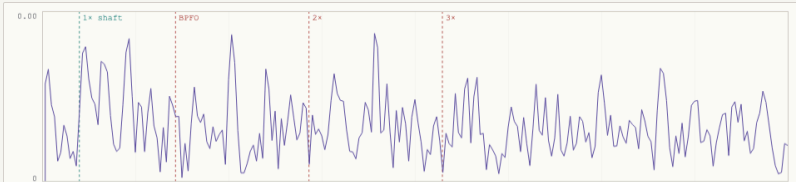
ACCELERATION DEFECT IMPACT INSTANTS

2 - RAW SPECTRUM (MAGNITUDE FFT, HANN WINDOW)



SPECTRUM SHAFT 1x AND HARMONICS STRUCTURAL RESONANCE BAND

3 - ENVELOPE SPECTRUM (BAND-PASS 2.4-4.0 KHz -> HILBERT -> FFT)



CLICK "HEALTHY" - FIRST THING AFTER THE ORIENTATION LINE.

WHAT LANDS ON SCREEN

- Annunciator: **Normal** lamp (teal)
- Bearing: no red spall visible
- Time waveform: clean-ish noise, no impact spikes
- Raw spectrum: dominant line at 30 Hz (shaft) + harmonics
- Envelope spectrum: **grass** - no BPFO peak
- Sliders: severity 0%, noise 18%, imbalance 35%

WHAT YOU SAY (SCRIPT §3.2)

Severity zero. Look at the raw spectrum: one dominant line at 30 Hz, the shaft, plus harmonics. Textbook. Envelope spectrum at the bottom is grass. Nothing at 107 Hz.

Kurtosis is low, the BPFO index sits around **0.3**, annunciator reads Normal.

NEXT

Click **EARLY DEFECT** — this is the moment the whole demo pivots on.

2 · Early defect — the key moment script §3.3 · 2.5 min

CLICK : EARLY DEFECT

Bearing Fault Lab

SKF 6205 deep-groove ball bearing, 9 rolling elements. Raise the outer-race defect and watch where it shows up: barely in the raw spectrum, unmistakably in the envelope spectrum.

Normal Alert Fault

BEARING - OUTER RACE DEFECT AT 12 O'CLOCK

DEFECT PASS RATE = BPFO = 107.4 Hz ORBIT SLOWED 17x

ROLLING ELEMENT DEFECT STRIKE

OPERATING CONDITIONS

SHAFT SPEED 1797 rpm

DEFECT SEVERITY 22 %

BACKGROUND NOISE 18 %

SHAFT IMBALANCE 35 %

HEALTHY **EARLY DEFECT** ADVANCED

NOISY PLANT

1 - TIME WAVEFORM, RAW ACCELEROMETER ($F_s = 20\,480\text{ Hz}$)

ACCELERATION DEFECT IMPACT INSTANTS

2 - RAW SPECTRUM (MAGNITUDE FFT, HANN WINDOW)

SPECTRUM SHAFT 1x AND HARMONICS STRUCTURAL RESONANCE BAND

3 - ENVELOPE SPECTRUM (BAND-PASS 2.4-4.0 KHz → HILBERT → FFT)

1x shaft BPFO 2x 3x

WHAT LANDS ON SCREEN

- Annunciator: **Alert** lamp (amber)
- Bearing: **red spall visible at 12 o'clock**
- Time waveform: modulated envelope, tiny impact ripples
- Raw spectrum: shaft line still dominates — **defect invisible**
- Envelope spectrum: **clean peak at 107.4 Hz + harmonics at 215, 322**
- Severity 22%, noise 18%, imbalance 35%

WHAT YOU SAY (SCRIPT §3.3) - PEDAGOGICAL MOMENT, DO NOT RUSH

Now there is a small spall. Watch the bearing: balls hitting the defect.

Look at the raw spectrum first. [point] Tell me where the fault is.

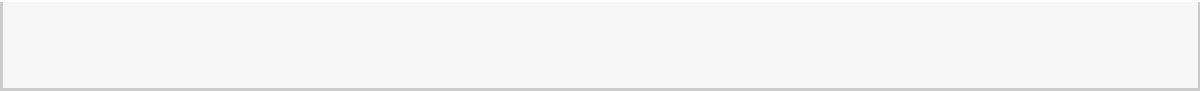
[Genuinely wait for an answer. Let them fail to find it.]

You cannot see it. The shaft line dominates completely. There is a slight lift in the resonance band around 3 kHz, and if you did not know to look there you would not notice. This is what a real early-stage bearing fault looks like on a raw FFT. Invisible.

Now the envelope spectrum. [point] There. A line at 107 Hz, and harmonics at 215 and 322. Exactly where our geometry said it would be. The BPFO index at the bottom right has gone from about **0.3 to about 3**, and the annunciator has moved to Alert.

Same data. Same measurement. The only difference is that we processed it in a way informed by the physics.

NEXT



Click **ADVANCED** to show what "too late" looks like. Skip §3.4 (change-the-speed) if running long.

Bearing Fault Lab

SKF 6205 deep-groove ball bearing, 9 rolling elements. Raise the outer-race defect and watch where it shows up: barely in the raw spectrum, unmistakably in the envelope spectrum.

Normal Alert Fault

BEARING - OUTER RACE DEFECT AT 12 O'CLOCK



DEFECT PASS RATE = BPFO = 107.4 Hz ORBIT SLOWED 17x

OPERATING CONDITIONS

SHAFT SPEED 1797 rpm

DEFECT SEVERITY 78 %

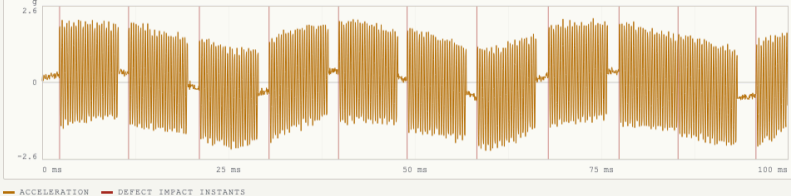
BACKGROUND NOISE 18 %

SHAFT IMBALANCE 35 %

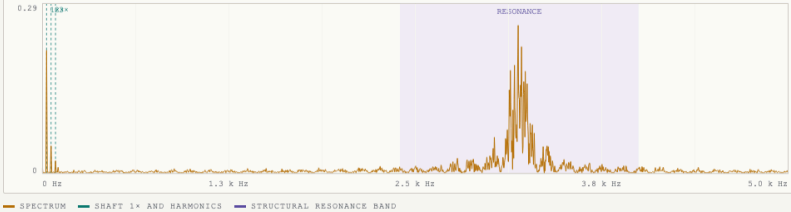
HEALTHY EARLY DEFECT ADVANCED

NOISY PLANT

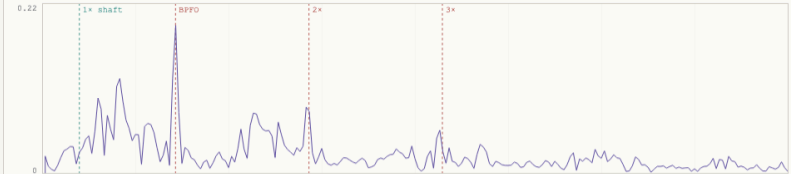
1 - TIME WAVEFORM, RAW ACCELEROMETER ($F_0 = 20\,480\text{ Hz}$)



2 - RAW SPECTRUM (MAGNITUDE FFT, HANN WINDOW)



3 - ENVELOPE SPECTRUM (BAND-PASS 2.4-4.0 KHz -> HILBERT -> FFT)



CLICK "ADVANCED" — SEVERITY JUMPS TO 78%.

WHAT LANDS ON SCREEN

- Annunciator: **Fault** lamp (red)
- Bearing: same red spall, larger visual halo
- Time waveform: **obvious periodic spikes** — no interpretation needed
- Raw spectrum: broad energy lift in the resonance band
- Envelope spectrum: BPFO peak is dominant
- Severity 78%, noise 18%, imbalance 35%

WHAT YOU SAY (SCRIPT §3.5 FIRST HALF)

Severity 78. Now it is obvious everywhere — visible in the raw waveform as periodic spikes, crest factor up, and the index around **7 to 8**. Fault lamp.

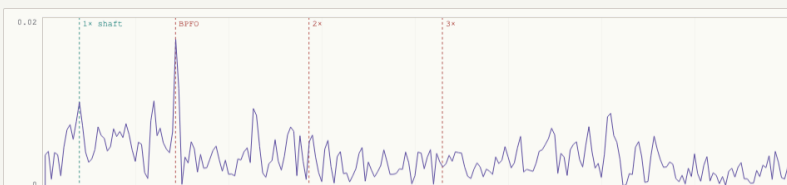
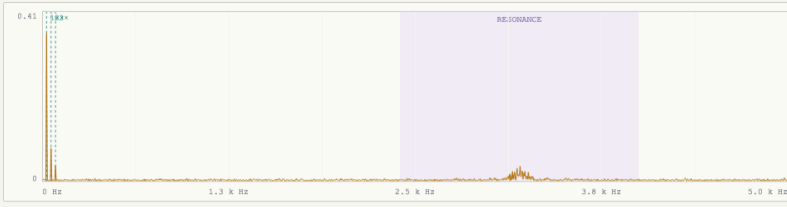
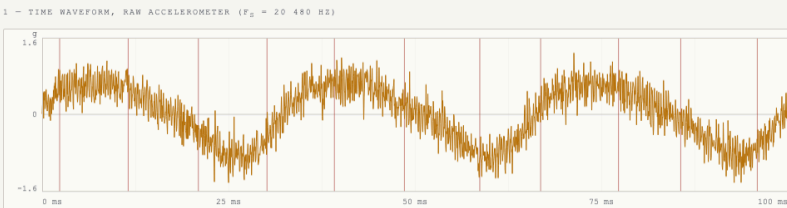
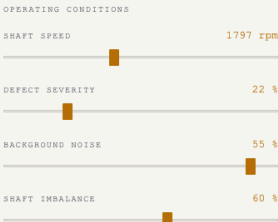
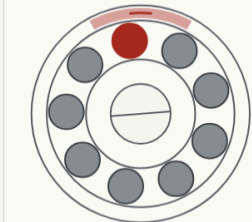
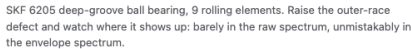
But notice: by the time it is this obvious, you have lost most of your warning time. **The value of the envelope method is entirely in the early case we just looked at.**

NEXT

Click **NOISY PLANT** — the honest one, and the pivot into the alarm-fatigue argument.

script §3.5 second half · 2 min

CLICK : NOISY PLANT



WHAT LANDS ON SCREEN

- Annunciator: **returns to Normal** (teal) — most of the time
- Bearing: same red spall — the defect has not gone anywhere
- Time waveform: noise dominates, impacts buried
- Raw spectrum: shaft harmonics broader, resonance smeared
- Envelope spectrum: **BPFO peak still there but much smaller**, buried in envelope grass
- Severity 22% (unchanged), noise 55%, imbalance 65%

WHAT YOU SAY (SCRIPT §3.5 SECOND HALF) - LET THE INDEX
SETTLE BEFORE SPEAKING

Last one, and this is the honest one. Same 22% defect as the early case — I have changed nothing about the bearing. All I have done is raise the plant noise and the shaft imbalance.

[Let them look at the index before you say the next part.]

The index has fallen from about 3 to about **1.6**, and look at the annunciator. It has gone back to **Normal**.

The defect is still there. It has not improved. Our detector has simply stopped seeing it. The alert threshold in this tool sits at 1.8, and on this setting the reading creeps above it maybe one time in eight — the same fault that was caught every single time in a quiet plant is now missed on most measurements.

That is a missed detection, and nobody gets told. A false alarm is annoying and visible. A missed detection is silent, and you only find out when the bearing fails.

NEXT

Move to Part 4 — Digital Twins. The alarm-fatigue arithmetic from slide 19 is what "the gap between 3 and 1.6" costs the plant.